AN EXPONENTIAL UPPER BOUND ON THE NUMBER OF BIPARTITIONS OF A SPARSE-BOUNDARY SET WITH FEW INTERNAL EDGES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned author source sources/combinatorics-fl04-vspace-1-5cm-perfect-matching-in-p/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. The author's displayed consequence labelled eq:colouring-weight at source lines 1360-1362 is delivered verbatim inside the statement span, so its label resolves inside the assembled material and no stub is needed for it.

Editorial note. Two displays inside the counted proof carry the author's labels eq:counting-lem and eq:size-X; they are inside the delivered span and are used by the proof itself.

Editorial note. The proof cites nothing, so the card needs no navigation stub; the bibtex database is declared because the source's own bibliography command is the only bibliography the paper has.

Allowed standard prerequisites (editorial).

- The graph G with its dimension-regularity parameters, the constants M and the relation ln n >= (ln 2/(M-1)) d. Set up in the author's introduction and preliminaries (source lines 201-240); used at the end of the counted proof. Supplies the parameter relation that turns the exponential factor 2^{|R|} (1-p_-)^{2(M-1)|R|} into exp(-Omega(|R|)).

Lemma 0.1. *Fix constants $K, K' > 0$. Let $R \subseteq V(G)$ have size $|R| \geq d$ and*

$$e_G(R, R^c) \leq K|R| \ln d.$$

The number of bipartitions $R = R_1 \sqcup R_2$ satisfying

$$e_G(R_1) + e_G(R_2) \leq K'|R|$$

is at most

$$\exp\left\{O\left(\frac{|R| \ln^2 d}{d}\right)\right\}.$$

Consequently, for every fixed $\theta > 0$,

$$\sum_{R=R_1 \sqcup R_2} (1-p_-)^{\theta(e_G(R_1)+e_G(R_2))} \leq \exp\{-o(|R|)\}. \tag{1}$$

Proof. Put $H = G[R]$ and $m = e_G(R_1) + e_G(R_2)$, and assume w.l.o.g. that $|R_2| \geq |R_1|$. Note that

$$e_G(R, R^c) = d|R| - \sum_{v \in R} d_H(v), \qquad 2m = \sum_{v \in R_1} d_{H[R_1]}(v) + \sum_{v \in R_2} d_{H[R_2]}(v). \tag{2}$$

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Let X be the set of vertices $v \in R$ with either few neighbours in H or many neighbours in their own bipartition class, i.e.,

$$X \coloneqq \left\{ v \in R : d_H(v) < \frac{3}{4}d \qquad \text{or} \qquad d_{H[R_i]}(v) > d/4 \quad (\text{ if } v \in R_i) \right\}.$$

By definition every $v \in R \setminus V$ has at least $\frac{d}{2}$ neighbours in the opposite bipartition class. From (2) we conclude that

$$|X| \leq \frac{4e_G(R, R^c)}{d} + \frac{8m}{d}. \tag{3}$$

Form a random set T by choosing each vertex of R_2 independently with probability $\rho = 12 \ln d/d$. The expected number of vertices of $R_1 \setminus X$ with no neighbour in T is at most

$$s(1-\rho)^{d/2} \leq \frac{|R|}{d^6}.$$

Moreover,

$$\mathbb{E}|T| = \rho|R_2| \leq \rho|R|, \qquad \mathbb{E}|R_2 \cap N_H(T)| \leq \rho 2e_G(R_2) \leq 2\rho m.$$

Applying Markov's inequality to these three quantities gives a set $T \subseteq R_2$ with

$$|T| \leq 3\rho|R|, \qquad |R_1 \setminus N_H(T)| \leq |X| + \frac{|R|}{d^2}, \qquad |R_2 \cap N_H(T)| \leq 6\rho m.$$

Combining (3) with our assumptions on $e_G(R, R^c)$ and m, we obtain that $R_1 \triangle N_H(T)$ has size $O(|R| \ln d/d)$. The pair $(T, R_1 \triangle N_H(T))$ determines R_1, and the number of possible pairs is at most

$$\left(\sum_{j \leq O(|R| \ln d/d)} \binom{|R|}{j} \right)^2 = \exp\left\{ O\left(\frac{|R| \ln^2 d}{d} \right) \right\},$$

which proves the first assertion.

For (1) apply the first assertion for the bipartitions with $m \leq \frac{2(M-1)}{\Theta}|R|$ to obtain that these contribute at most $\exp\{o(|R|)\}$. For the bipartitions with $m > \frac{2(M-1)}{\Theta}|R|$ contribute at most

$$2^{|R|}(1-p_-)^{2(M-1)|R|} = \exp\left(-\Omega(|R|)\right),$$

using again that $\ln n \geq \frac{\ln 2}{M-1} d$. □